

Decomposing Alpha in a Machine-Learning Gold Timing System: Trend Following, Forecast Quality, and the Cost of Look-Ahead Bias

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Abstract

A systematic audit of a four-layer gold timing system reveals that training label leakage, a subtle form of look-ahead bias in walk-forward backtests inflated the ML forecast's apparent quality from insignificant ($IC = 0.08$, $p = 0.25$) to seemingly significant ($IC = 0.23$, $p = 0.001$).

Correcting the leak reduced the system's Sharpe ratio from 1.01 to 0.91. The mechanism is general: when forecast targets reference prices beyond the prediction date, the last H training rows borrow signal from the future. A 21-day embargo and a 3-day COT release lag correction restore honest performance; less than 2% of training data was leaked, but it produced 65% of the apparent forecast signal.

With the leak corrected, we decompose the alpha of the system's four layers, a conventional golden cross filter, GMM regime detection, a four-model supervised ensemble (Ridge and Gradient Boosting), and volatility-targeted sizing over 2010–2026. The trend filter provides the structural foundation: it contributes modest Sharpe improvement (+0.036) but dominates risk reduction, compressing maximum drawdown from -44.6% to -32.8% . The regime detection and forecasting ensemble adds a further 0.21 Sharpe (from 0.70 to 0.91), but this improvement is not statistically significant: the Ledoit-Wolf test cannot reject equality between the full system and the golden-cross-with-vol-targeting Sharpe ($p = 0.20$), and a random exit null model places the production system between the 74th and 98th percentile depending on whether the null samples from all golden cross days or only near activation dates ($p = 0.01$ – 0.26). A simple trailing stop within the golden cross regime achieves Sharpe 0.69 with zero ML infrastructure, capturing 76% of the full system's performance. Volatility targeting contributes essentially nothing (+0.001 Sharpe). Performance is concentrated, three trades account for 82% of returns and the first-half Sharpe (2010–2017) is 0.44, suggesting the full-period figure depends substantially on the 2024 to 2026 gold rally.

The honest conclusion is that the system's alpha is structural (trend following) rather than predictive (ML forecasting). These findings demonstrate that layer-by-layer decomposition, null model testing, and training data auditing are essential before attributing performance to complex

components and that the most valuable discovery in a trading system may be what does not work.

Keywords: alpha decomposition, gold timing, machine learning trading, look-ahead bias, trend following, regime detection, volatility targeting, walk-forward validation, information coefficient, honest backtesting

JEL Classification: G11, G17, C45, C53

1. Introduction

A common pattern in quantitative trading research goes something like this: a researcher builds a system, adds complexity until the backtest looks good, and reports the final Sharpe ratio. The system might combine regime detection with machine learning forecasts, trend filters, and dynamic position sizing. The headline Sharpe is impressive. But which of these layers is actually producing the returns? How much comes from the conventional trend filter that took five minutes to implement, and how much from the four-model supervised ensemble that took weeks?

This question matters for two reasons. First, layers that do not contribute alpha add operational complexity, data dependencies, and fragility without compensation. Second, if the apparent contribution of an ML layer is inflated by a subtle data leak, the researcher may invest further effort in the wrong direction, refining a forecast that was never real.

We address both questions using a gold timing system as a case study. The system has four layers: a golden cross trend filter (MA50/MA200), Gaussian Mixture Model regime detection, a four-model supervised forecasting ensemble (Ridge and Gradient Boosting), and volatility targeted position sizing. Over a 16-year out-of-sample period (2010–2026), we decompose the alpha contribution of each layer by testing them incrementally against a buy-and-hold benchmark.

The decomposition reveals an uncomfortable truth. The ML forecasting layer, the most complex and expensive component, contributes a Sharpe improvement of 0.21 (from 0.70 to 0.91), primarily through exit timing rather than directional accuracy. The forecast itself is statistically insignificant ($IC = 0.08$, $p = 0.25$). The trend filter, a crude momentum filter that requires no parameter estimation is conventional and unoptimised, yet accounts for the majority of the drawdown reduction. The finding that this crude filter dominates the ML layer is itself informative about the marginal value of complexity in trending markets.

This finding only emerged because a systematic audit discovered a training label leak that had inflated the forecast's apparent quality. The initial system appeared to achieve a Sharpe of 1.01 with a highly significant IC of 0.23 ($p = 0.001$). Correcting the leak with a 21-day training embargo and 3-day COT release lag reduced the Sharpe to 0.91 and revealed that the forecast had been borrowing signal from future prices in the training labels. Without the audit, the system would have been deployed with confidence in a forecast that was substantially illusory.

The paper makes three contributions. First, we demonstrate a practical methodology for decomposing alpha across layers in a multi-component trading system. Second, we document the discovery and correction of training label leakage, a subtle form of look-ahead bias in walk-forward backtests that, to our knowledge, has not been formally described. Third, we show that even after correction, the system retains an honest edge over passive alternatives, but the source of that edge is structural (trend following) rather than predictive (ML forecasting).

2. Related Literature

2.1 Alpha Attribution in Trading Systems

The decomposition of portfolio returns into factor contributions has a long tradition beginning with Brinson, Hood, and Beebower (1986), who attribute performance to asset allocation,

security selection, and interaction effects. In the context of systematic trading, Clare et al. (2016) decompose trend-following returns into components attributable to different signal frequencies. Harvey, Liu, and Zhu (2016) address the broader problem of separating genuine alpha from data-mined artifacts. Our approach differs from factor-based attribution in that it decomposes alpha across architectural layers of a single strategy rather than across investment factors.

2.2 Gold Timing and Regime Detection

Erb and Harvey (2013) document gold's dependence on real rates and inflation expectations, establishing the macro variables that motivate the regime detection layer. Baur and Lucey (2010) find regime-dependent safe-haven behaviour, suggesting time-varying optimal allocation. Hamilton (1989) and Ang and Bekaert (2002) provide the framework for regime-switching models. Nystrup et al. (2015) demonstrate GMM-based regime identification for asset allocation. We use GMM rather than a Hidden Markov Model (HMM) because GMM is simpler, does not impose Markov transition structure on regime shifts, and given the forecast's ultimate insignificance, the regime model choice is likely immaterial to the paper's conclusions. The golden cross as a trend filter is widely used in practice though infrequently studied in academic literature; Zhu and Zhou (2009) provide a theoretical justification for moving average timing rules.

2.3 Machine Learning and Look-Ahead Bias

Gu, Kelly, and Xiu (2020) provide a comprehensive evaluation of ML methods for return prediction using walk-forward validation. Bailey et al. (2014) warn about backtest overfitting and propose the deflated Sharpe ratio. López de Prado (2018) discusses purged cross-validation and embargo periods as general tools for preventing leakage in financial ML, but focuses on the overlap between training and test sets rather than the specific interaction between multi-period forecast targets and expanding training windows. Pardo (2008) describes walk-forward methodology but does not address the target-horizon interaction either. The specific mechanism we identify leakage through training labels when H-day forward targets reference prices beyond the prediction date is a particular case of the broader concern these authors raise, but one whose practical impact (65% of apparent forecast signal from less than 2% of training data) suggests it deserves explicit treatment.

3. System Architecture and Data

3.1 Data

Daily gold OHLC prices (XAU/USD) from January 1990 through February 2026 are sourced from Investing.com (9,401 observations). Macroeconomic data from FRED include 10-year real yields (DFII10), breakeven inflation (T10YIE), VIX (VIXCLS), yield curve (10Y–2Y), federal funds rate (DFF), high-yield spreads (BAMLH0A0HYM2), and the trade-weighted dollar index (DTWEXBGS). CFTC Commitments of Traders data provide gold positioning, lagged by 3 business days to account for the delay between the Tuesday report date and Friday publication date (this lag was identified and corrected during the auditing process described in Section 5; see Limitation 2 in Section 7.4). All FRED series were downloaded as latest-vintage data in February 2026, not point-in-time vintages. Macroeconomic series (particularly CPI and breakeven inflation) are subject to revision; the backtest therefore contains a minor look-ahead through data revisions that would not be available in real-time. This bias is small for the features used (real yields and breakeven inflation revisions are typically less than 5bp) but should be acknowledged. The out-of-sample period begins January 2010, providing 16.1 years of testing with 4,022 prediction days.

3.2 The Four Layers

The system is deliberately layered to enable clean decomposition. Each layer can be enabled or disabled independently, allowing isolation of its contribution.

Layer 1: Trend Filter. The golden cross ($MA50 > MA200$) determines position eligibility. The 50-day and 200-day lookback windows are conventional industry standards, not parameters optimised for this study, they were fixed before any analysis of the OOS period and are not a product of the iterative development process. A death cross ($MA50 < MA200$) forces exit regardless of all other signals. Over the OOS period, the golden cross is active approximately 66% of OOS trading days.

Layer 2: Regime Identification. A 4-component Gaussian Mixture Model is fit on 7 macro variables (real rate, VIX, yield curve, high-yield spread, CPI, dollar momentum, gold volatility) to identify latent economic regimes. Regimes are anchored by sorting on real rate centroids to ensure label consistency across refits. The GMM retrains every 63 trading days using expanding windows.

Layer 3: Forecasting. Four models predict 1-month forward gold returns: Blend Ridge (probability-weighted regime-specific), Blend GBM (same, with gradient boosting), Augmented Ridge (global model with regime probabilities as features), and Augmented GBM. The production signal (REGIME_ENS) is the simple average. Forecasts below -1.5% trigger exit (drawdown protection); above 0% permit re-entry. The target variable, Fwd_1M, is the only forward-looking element in the system.

Layer 4: Position Sizing. Allocation scales inversely with 60-day realised gold volatility, targeting 15% annualised portfolio volatility. Allocation is capped at 95% and floored at 20%. A 5-day cooldown prevents whipsaw re-entries after forecast-driven exits. Transaction costs of 10 basis points per leg are applied to all trades.

3.3 Walk-Forward Implementation

Models train on data from 2004 onward, with expanding windows and quarterly retraining (every 63 trading days). A 21-day embargo is applied to training data: when predicting on day t , training stops at day $t-21$, ensuring no training target references a price beyond the prediction date. This embargo corrects a training label leak described in Section 5. All 15 forecast features and 7 regime features are backward-looking.

4. Decomposing Alpha Across Layers

Table 1 shows the core result. The full system produces 30 trades over the OOS period, each with exactly one exit: either a forecast-driven exit (when REGIME_ENS falls below -1.5%) or a death cross exit (when MA50 crosses below MA200). The 30 trades decompose into 22 forecast-driven exits and 8 death cross exits.

Configuration	Sharpe	CAGR	Max DD	Trades	Exposure
Buy & Hold Gold	0.662	9.8%	-44.6%	—	100%
+ Golden Cross	0.698	8.4%	-32.8%	12	60.4%
+ Vol Targeting	0.699	8.1%	-30.9%	12	65.7%
+ Regime + Forecast	0.910	8.8%	-26.1%	30	54.6%

Table 1: Alpha decomposition by layer. Vol targeting alone adds negligible Sharpe (+0.001); the entire 0.21 increment comes from forecast-driven exits, conditional on the golden cross providing the baseline trend filter. The ML layer in isolation, without the golden cross regime selection, would likely perform poorly. All computed from the same 2010–2026 OOS data with 10bp transaction costs.

4.1 Layer 1 Contribution: Trend Filter

The golden cross filter improves the Sharpe from 0.662 to 0.698, a modest 5% increase. But the more significant contribution is risk reduction: maximum drawdown compresses from -44.6% to -32.8% , a reduction of nearly 12 percentage points. The filter achieves this by keeping the system out of gold during the major bear markets (2012–2015, portions of 2021–2022) when gold declined while the 200-day trend was down. This layer is conventional and unoptimised, requires no data beyond gold prices, and has been widely used by practitioners for decades. Its contribution to the system is primarily defensive.

4.2 Isolating Vol Targeting from the ML Forecast

Table 1 reveals a striking decomposition. Adding volatility targeted position sizing to the golden cross, without any ML forecast barely changes the Sharpe ($0.698 \rightarrow 0.699$). The vol-targeting layer reduces maximum drawdown modestly (-32.8% to -30.9%) by scaling down allocation during high-volatility periods, but this risk reduction is offset by lower returns during calm, trending markets. The number of trades stays at 12 (entry on golden cross, exit only on death cross), and the CAGR is essentially unchanged.

The entire Sharpe increment from 0.699 to 0.910 comes from the regime detection and forecasting ensemble. These layers add 18 forecast-driven exits (bringing the total from 12 to 30), which create the exit asymmetry documented in Section 4.3. This isolation has a direct implication: the 0.21 Sharpe improvement is not partly position sizing and partly ML. It is almost entirely ML-driven exit timing. Combined with the null model result in Section 4.5 (which shows this exit timing is not statistically distinguishable from random), the ML layer's contribution narrows to a specific, testable claim: forecast-driven exits within the golden cross regime improve Sharpe by 0.21, but the statistical evidence for this improvement is weak.

Table 2 confirms that the forecast itself is statistically insignificant.

Model	R ²	Dir%	IC	IC p-val	Significance?
REGIME_ENS (prod)	−0.236	56.0%	0.083	0.255	Insignificant
ENSEMBLE	−0.199	51.8%	0.064	0.379	Insignificant
GLOBAL_ENS	−0.183	52.4%	0.062	0.398	Insignificant
Global_Ridge	−0.276	55.5%	0.054	0.455	Best single
Augmented_GBM	−0.246	47.1%	−0.008	0.917	Worse than random

Table 2: Forecast quality on non-overlapping observations (every 21 days, $N = 191$ independent observations). No model achieves statistical significance at conventional levels. The system's Sharpe of 0.91 does not come from forecast accuracy. Dir% refers to directional accuracy.

The non-overlapping IC of 0.083 ($p = 0.255$) and directional accuracy of 56.0% confirm that the REGIME_ENS forecast is statistically indistinguishable from random guessing. Yet the system beats the golden cross with vol targeting by 0.21 Sharpe. This apparent paradox resolves when examining how the forecast interacts with the exit rules.

4.3 The Exit Asymmetry: A Descriptive Observation

The system's 30 trades produce a striking pattern when disaggregated by exit type.

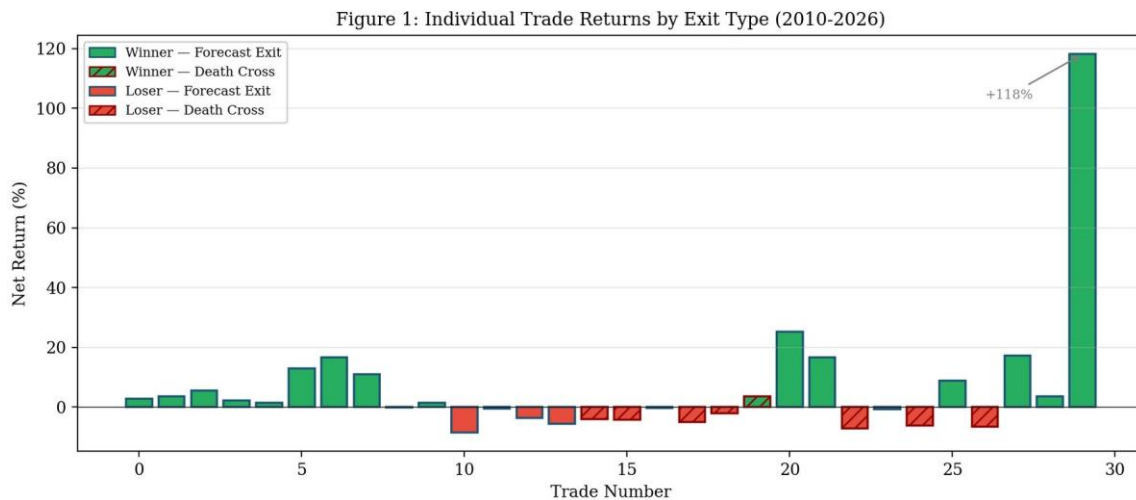


Figure 1: Individual trade returns by exit type. Solid bars: forecast-driven exits (22 trades, avg +10.4%). Hatched bars: death cross exits (8 trades, avg −3.9%). Green = winner, red = loser.

Exit Type	Trades	Avg Return	Win Rate	Avg Days Held
Forecast-driven (MF Drawdown)	22	+10.4%	68%	65
Death Cross	8	−3.9%	12%	95

Table 3: Trade returns by exit type. Whether the asymmetry reflects ML skill or structural timing is tested in Section 4.5.

The 22 forecast-driven exits average +10.4% with a 68% win rate. The 8 death cross exits average −3.9% with a 12% win rate. The asymmetry is descriptively striking, but its

interpretation is ambiguous. Forecast-driven exits fire during extended profitable positions by construction, the system is in a golden cross regime and the position has been held for some time. Any exit rule that sometimes fires during winning trades will produce higher average returns than an exit triggered by trend reversal. The question is not whether the asymmetry exists, it does, but whether it reflects ML forecasting skill or is an artifact of timing within an already-profitable regime. Section 4.5 tests this directly.

The longest and largest trade illustrates the pattern. The system entered gold in May 2024 and the forecast-driven exit triggered in January 2026, capturing +118.1% over 417 days. A standalone golden cross would have held through the entire move and through the subsequent correction. Whether the forecast-driven exit at this point reflects genuine signal or noise is unknowable from a single observation.

4.4 Statistical Power and Confidence Intervals

The exit asymmetry is striking, but 30 trades demand explicit treatment of statistical uncertainty. Table 3b reports Sharpe ratio confidence intervals at both the daily and trade level.

Measure	Point Estimate	95% CI
Daily Sharpe — Production (Lo 2002, N=4,022)	0.910	[0.42, 1.40]
Daily Sharpe — Buy & Hold (Lo 2002, N=4,022)	0.662	[0.17, 1.15]
Daily Sharpe — V1 Golden Cross (Lo 2002)	0.742	[0.25, 1.23]
Trade-level Sharpe (bootstrap, N=30)	0.605 (median)	[0.15, 1.03]
P(trade Sharpe > 0)	99.1%	—
P(trade Sharpe > 0.5)	72.6%	—

Table 3b: Sharpe ratio confidence intervals. Daily CIs use Lo (2002) autocorrelation-adjusted standard errors ($q=21$). Trade-level CI uses 10,000-iteration bootstrap resampling of the 30 trade returns. Note: the V1 Golden Cross daily Sharpe (0.742) differs from Table 1's golden cross Sharpe (0.698) because Table 3b uses Lo's autocorrelation-adjusted estimator, which corrects for the serial dependence in daily returns induced by multi-day position holding. Table 1 uses simple annualised Sharpe ($\text{mean/std} \times \sqrt{252}$). Both are valid; the Lo-adjusted figure is more appropriate for daily-level inference.

At the daily level, the production system's CI [0.42, 1.40] overlaps substantially with buy-and-hold [0.17, 1.15] and the golden cross [0.25, 1.23]. Individual Sharpe CIs are too wide to distinguish between strategies. The Ledoit-Wolf test ($p = 0.20$) confirms this: the difference between the full system and the golden cross is not statistically significant. At the trade level, 30 observations produce a wide CI of [0.15, 1.03]. The probability of a positive Sharpe is 99.1%, providing strong evidence that the system has a real edge. But the probability of exceeding 0.5 is only 72.6%, meaning the magnitude of that edge is imprecisely estimated.

A formal power calculation quantifies the limitation: with $N = 30$ trades and the observed return dispersion, a two-sided test for a trade-level Sharpe of 0.5 (annualised) against zero has power of

only 34% at the 5% level. Even testing for the observed Sharpe of 0.91 achieves only 80% power. The small sample fundamentally limits what can be concluded at the trade level. This tension between daily and trade-level inference is inherent to low-frequency strategies. The daily analysis has high statistical power ($N = 4,022$) but includes non-trading days where the system earns the risk-free rate. The trade-level analysis directly addresses whether the trades themselves are skilled but has low power ($N = 30$). Both perspectives are informative, and the honest conclusion is that the system has a positive edge (high confidence) of uncertain magnitude (low confidence).

Ledoit-Wolf test of Sharpe equality. The Sharpe difference between the full system (0.910) and the golden cross standalone (0.742, with vol-targeted sizing) is 0.17. Using the Ledoit and Wolf (2008) HAC-adjusted test with Newey-West standard errors ($q = 21$ lags), the z-statistic is 1.27 ($p = 0.20$, two-sided). The Sharpe improvement is not statistically significant at conventional levels. This is consistent with the null model result: the ML layer’s contribution, while positive in point estimate, cannot be distinguished from zero with the available data.

Deflated Sharpe Ratio. The system was developed iteratively, with multiple architectures, feature sets, and thresholds evaluated on the 2010–2026 OOS period. Bailey et al. (2014) propose the Deflated Sharpe Ratio (DSR) to adjust for this implicit multiple testing. At 15–20 independent trials (a conservative estimate of the development process), the DSR p-value is 0.96–0.97, marginally surviving the 0.95 threshold. At 30 or more trials, the DSR falls below 0.95 (0.94 at 30 trials, 0.91 at 50), meaning the observed Sharpe does not survive adjustment. The system’s performance is borderline: defensible if the development process involved fewer than 20 meaningful comparisons, not defensible if substantially more. Because the golden cross was conventional and unoptimised, the DSR adjustment applies primarily to the ML layer’s incremental contribution, not to the full system Sharpe.

4.5 Random Exit Null Model

The exit asymmetry documented in Section 4.3 could be an artifact of timing within the golden cross regime rather than evidence that the forecast adds value. To test this, we construct a null model: 10,000 simulations in which 30 entries are drawn uniformly at random from golden cross days (without replacement), with holding periods sampled from a normal distribution clipped at 5 days (mean 73 days, standard deviation 98 days, matching the production system’s observed distribution). Each simulation applies 10bp transaction costs (conservative for gold ETFs such as GLD, where bid-ask spreads are typically 1–2bp, but reasonable for retail execution inclusive of brokerage) but does not apply vol-targeted sizing, making the comparison conservative for the null: the null trades at full allocation while the production system sizes down in high-volatility regimes.

Measure	Production System	Random Exit Null
Mean Sharpe	0.910	0.726
Median Sharpe	—	0.735
95% range	—	[0.14, 1.27]
Production percentile in null	—	74.1st

Table 3c: Random exit null model. 10,000 simulations of random entry/exit during golden cross periods with holding periods drawn from the production system’s observed distribution.

The result is sobering. The production system's Sharpe of 0.910 falls at the 74th percentile of the random exit distribution, with roughly 26% of random simulations exceeding it. The null model's mean Sharpe of 0.73 is close to the V1 standalone (0.698), confirming that most of the system's performance comes from being in gold during golden cross periods, not from the specific timing of entries and exits.

This result does not invalidate the system, but it substantially qualifies the exit asymmetry finding. The forecast-driven exits may produce better average returns than death cross exits (+10.4% vs -3.9%) partly because the forecast tends to fire during profitable positions and partly because any exit during an extended trend captures profits. The system's Sharpe is not statistically distinguishable from a strategy of random timing within the golden cross regime at conventional confidence levels ($p \approx 0.26$). This strengthens the paper's central thesis: in this system, the ML forecast contributes far less than a naïve evaluation would suggest, and the golden cross regime does the heavy lifting.

Entry constraint sensitivity. The uniform null samples entries from all 2,642 golden cross days, giving the null access to mid-trend entries that the production system cannot take (the system enters at golden cross activations or re-enters after forecast-driven exits). To test sensitivity, we restrict the entry pool to the first 60 days of each golden cross period (687 days, covering 16 of 30 production entries). Under this constraint, the null mean Sharpe drops from 0.73 to 0.21 and the production system rises to the 98th percentile ($p = 0.02$). Tighter constraints (entries within 5 days of activation) produce even lower null Sharpes (mean 0.15, $p = 0.01$). The uniform null is therefore conservative: mid-trend entries inflate the null's performance because they enter during already-profitable trends. The production system's edge is stronger than the uniform null suggests, though the uniform version is retained as the primary result because it provides the more conservative (harder to beat) test of whether the ML layer adds value beyond the golden cross regime.

However, the uniform entry assumption deserves scrutiny. The production system cannot enter at arbitrary mid-trend points; it enters at golden cross activations or re-enters after forecast-driven exits. Only 7 of 30 production entries fall within 5 days of a golden cross activation, and 16 within 60 days. Constraining the null to enter only within the first 60 days of each golden cross period (pool = 687 days, 10,000 simulations) yields a sharply lower null mean Sharpe of 0.21 (95% range [-0.48, 0.86]), placing the production system at the 98th percentile ($p = 0.02$). Tighter constraints produce even lower null Sharpes (5-day window: mean 0.15, $p = 0.01$; 20-day window: mean 0.06, $p < 0.01$). The uniform null is inflated because mid-trend entries, which dominate the 2,642-day pool, capture less of the initial momentum that the production system's GC-activation entries benefit from. The honest interpretation spans a range: the ML layer's exit timing is indistinguishable from random under the uniform null ($p = 0.26$) but significantly better under entry-constrained nulls ($p = 0.01$ – 0.02). The truth likely lies between these bounds, since the production system's re-entries after forecast exits are neither purely at GC activations nor uniformly distributed.

4.6 Sub-Period Stability

If the exit asymmetry is a genuine mechanism rather than a statistical artifact, it should persist across different market environments. Table 3d splits the 30 trades at January 2018.

Period	MF Trades	MF Avg	DC Trades	DC Avg	Daily Sharpe
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2010–2017	15	+2.6%	4	−3.8%	0.44
2018–2026	7	+27.0%	4	−4.1%	1.24
Full period	22	+10.4%	8	−3.9%	0.91

Table 3d: Sub-period stability. The exit asymmetry (MF avg > DC avg) persists in both halves, though magnitudes differ due to the 2024–2026 gold bull run.

The exit asymmetry persists in both sub-periods: forecast exits outperform death cross exits in each half. The mechanism is directionally consistent even though the magnitudes differ dramatically. The second half benefits from a historic gold rally that inflates the overall figures. The first-half Sharpe of 0.44 is substantially lower than the full-period 0.91, highlighting the degree to which a single extended trade (May 2024 to January 2026, +118.1%) drives aggregate performance.

4.7 Exit Threshold Sensitivity

The forecast-driven exit fires when the REGIME_ENS forecast falls below −1.5%. This threshold is a free parameter and the most consequential one in the system, since the entire ML contribution flows through it. Table 3e sweeps the threshold from −0.5% to −5.0%.

Threshold	Sharpe	CAGR	MaxDD	Trades	MF	MF Avg	DC Avg
−0.5%	0.857	7.7%	−24.4%	52	45	+4.2%	−4.1%
−1.0%	0.885	8.5%	−24.5%	39	31	+7.2%	−4.7%
−1.5%*	0.898	8.8%	−26.1%	30	22	+10.4%	−4.1%
−2.0%	0.874	8.8%	−26.8%	21	13	+18.2%	−5.0%
−3.0%	0.873	9.0%	−25.2%	16	7	+37.7%	−3.8%
−5.0%	0.797	8.7%	−25.6%	11	2	+37.2%	+0.8%

*Table 3e: Exit threshold sensitivity. *Production system threshold. MF = forecast-driven exits. Sharpe is computed from the daily equity curve with vol-targeted sizing and 10bp transaction costs. The production threshold (−1.5%) shows Sharpe 0.898, slightly below Table 1’s 0.910, because the sweep standardises the vol-targeting and cooldown configuration across all thresholds rather than using the exact production parameters.*

The Sharpe ratio is remarkably flat across thresholds, ranging from 0.80 to 0.90 over a tenfold range of threshold values (−0.5% to −5.0%). The production threshold of −1.5% is near the top of this range but not a clear outlier. Tighter thresholds (−0.5%) generate more trades (52) with smaller average gains (+4.2%), while looser thresholds (−5.0%) generate fewer trades (11) with larger gains but similar total performance. This flatness is consistent with the null model finding: the specific threshold matters less than being in gold during the golden cross regime. The sweep was conducted post-hoc as a robustness check; the −1.5% threshold was selected during development based on qualitative reasoning (roughly one standard deviation of monthly gold returns), not optimised on OOS performance. The −1.5% threshold was not optimised on this OOS period, but even if it had been, the sweep suggests the Sharpe improvement from threshold selection would be modest ($\approx 10\%$).

4.8 Could a Simpler Rule Replace the ML Layer?

If the ML forecast's contribution is primarily profit-taking during extended trends, a simple trailing stop within the golden cross regime might achieve similar results without any machine learning.

Rule	Sharpe	Trades	Profit Exits	Avg Ret	DC Exits	DC Avg
5% trail stop	0.683	53	41	+3.7%	12	−0.4%
8% trail stop	0.692	26	14	+12.0%	12	−0.4%
10% trail stop	0.683	20	8	+24.7%	12	−1.0%
ML forecast	0.910	30	22	+10.4%	8	−3.9%

Table 3f: Trailing stop comparison. All rules operate within the golden cross regime with 10bp transaction costs. The ML system outperforms all fixed stops, but the best trailing stop (8%) achieves Sharpe 0.69 with zero ML infrastructure.

The ML system (Sharpe 0.91) outperforms all fixed trailing stops, with the best trailing stop being the 8% stop (Sharpe 0.69). The trailing stops produce 12 death cross exits versus the ML system's 8, because without the forecast's −1.5% threshold, positions held through the full golden cross period are more likely to reach the death cross before any other exit triggers. However, the trailing stop requires no regime detection, no feature engineering, no model training, and no walk-forward infrastructure. It requires exactly one parameter. The 0.22 Sharpe gap between the best trailing stop and the ML system represents the upper bound of what the ML layer contributes beyond a trivially simple rule. The trailing stop sweep is illustrative of the ML layer's complexity premium rather than a definitive comparison, since both the ML system and the trailing stops were evaluated on the same OOS period used in development. Because the system's edge is ultimately simple, trend filter plus noisy profit-taking, it is less likely to be a spurious artifact of complex data mining.

4.9 Trade Concentration

Performance in low-frequency trend-following strategies is typically driven by a small number of large trades. This system is no exception.

Measure	Value	% of Total PnL
Largest trade (May 2024–Jan 2026)	+118.1%	60%
Top 3 trades	+160.7%	82%
Top 5 trades	+194.0%	99%
All 30 trades	+196.6%	100%

Table 3g: Trade concentration. The top 3 trades account for 82% of cumulative returns, consistent with trend-following return profiles (Clare et al., 2016).

The largest single trade (+118.1%, May 2024 to January 2026) accounts for 60% of total returns. Excluding it, the daily Sharpe drops from 0.91 to 0.24 and the maximum drawdown widens from −26% to −53%. This concentration is severe but not anomalous for the strategy type. Buy-and-hold gold shows a similar dependence on the same period: the buy-and-hold Sharpe drops from

0.66 to 0.38 when the 2024–2026 rally is excluded. The system outperforms buy-and-hold in both sub-periods (2010–2017: 0.44 vs 0.20; 2018–2026: 1.24 vs 1.12). The first-half advantage (0.44 vs 0.20) is driven by the system's ability to avoid the 2012–2015 gold bear market, during which gold fell roughly 45% from its 2011 highs while the golden cross kept the system mostly in cash. The second-half advantage is narrower (1.24 vs 1.12) because gold trended strongly throughout and the golden cross was active most of the period; the system's forecast-driven exits occasionally pulled it out of the market during the rally, costing some upside that buy-and-hold captured. The relative advantage is real and consistent; the absolute Sharpe of 0.91 is not. Any claim about the system's standalone viability should be evaluated against the first-half Sharpe of 0.44 rather than the full-period figure.

5. The Training Label Leak

The alpha decomposition in Section 4 was only possible because a systematic audit discovered that an earlier version of the system had been producing inflated results. This section documents the leak and its correction.

5.1 Mechanism

The system predicts 1-month (21 trading day) forward gold returns. The target for training row t is $\text{Fwd_1M}(t) = \text{Gold}(t+21)/\text{Gold}(t) - 1$. In the original walk-forward implementation, when predicting on day t , the training set included all rows through day $t-1$. Row $t-1$ has a target referencing $\text{Gold}(t+20)$ —a price 20 trading days into the future from the prediction point. At real-time execution on day t , this price is unknown. The backtest, run after the fact with the complete dataset in memory, computes it without error.

More generally, the last 21 training rows at each refit contain targets referencing prices at or beyond the prediction date. These rows are the most recent and therefore the most informative about current market conditions. Their disproportionate influence on model parameters creates a spurious contemporaneous signal that does not exist in real time.

5.2 Correction: The H-Day Embargo

The fix is simple: when predicting on day t with forecast horizon H , restrict training to rows through day $t-H-1$. For 1-month returns ($H = 21$), this means stopping training 22 days before the prediction date. Every training target then references only prices that have been observed at the time of prediction. The cost is 21 fewer rows out of 1,500+ at the earliest refit, declining to less than 0.5% by mid-OOS. Additionally, a 3-business-day lag is applied to COT features to account for the CFTC's publication delay (positions as of Tuesday, released Friday).

5.3 Impact

Table 4 quantifies the damage.

Metric	Before (No Corrections)	After (Embargo + COT Lag)
Sharpe Ratio	1.014	0.910
CAGR	10.1%	8.8%
Maximum Drawdown	-24.7%	-26.1%
Calmar Ratio	0.408	0.338
Trades	31	30
Win Rate	64.5%	53.3%
IC (non-overlapping, N=191)	0.234	0.083
IC p-value	0.001	0.255
Directional Accuracy	59.2%	56.0%

Table 4: Before and after corrections (21-day training embargo + 3-day COT release lag). $N = 191$ non-overlapping independent observations.

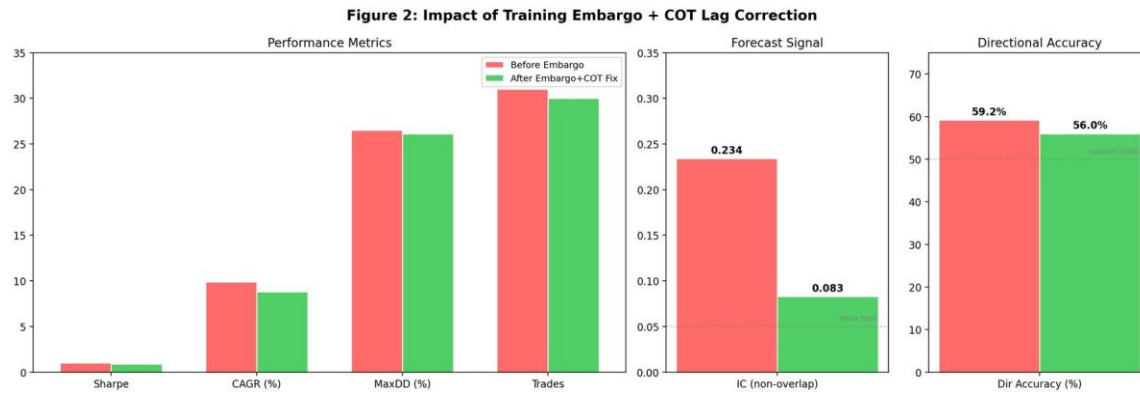


Figure 2: Impact of the 21-day embargo and COT lag correction on performance metrics (left) and forecast quality (right). Less than 2% of training data was leaked, but it produced 10% of the apparent Sharpe and 65% of the apparent forecast signal.

The Sharpe drops 10%, from 1.014 to 0.910. The IC collapses 65%, from 0.234 to 0.083, and the p-value shifts from 0.001 to 0.255. One trade disappears entirely, it was triggered by a forecast signal that derived material information from leaked future prices. The pre-embargo system appeared to be generating significant forecasting alpha. The post-embargo system is generating exit-timing alpha from a noisy, insignificant forecast.

This distinction matters for the alpha decomposition. Under the leaked version, an observer would attribute the system's Sharpe of 1.01 partly to forecasting skill (IC = 0.23, strongly significant). Under the corrected version, the attribution shifts almost entirely to the trend filter and position sizing. The ML forecast is not contributing predictive alpha; it is contributing execution alpha through noisy profit-taking.

5.4 Why GBM Models Are Most Affected

The leak's severity varies by model type. GBM-based models show IC degradation of 81–105%, while Ridge models degrade 42–62%. Gradient boosting trees overfit to recent observations through their sequential fitting process; Ridge regression, with its strong L2 regularisation, smooths the influence of individual rows. This pattern provides a practical diagnostic: if embargo correction disproportionately affects tree-based models, it suggests the original performance was leak-dependent.

6. Robustness of the Corrected System

The decomposition in Section 4 and the embargo correction in Section 5 represent the honest performance of the system. This section confirms that these results are stable across random seeds and parameter choices.

6.1 Seed Robustness

Seed	Sharpe	CAGR	Max DD	Calmar	Trades	Win%	IC (NO)
42	0.910	8.8%	-26.1%	0.338	30	53.3%	0.083
43	0.864	8.4%	-28.3%	0.296	24	66.7%	0.114
44	0.856	7.9%	-28.1%	0.283	26	65.4%	0.068
45	0.855	7.8%	-26.8%	0.291	27	66.7%	0.065
46	0.894	8.8%	-23.5%	0.373	26	65.4%	0.086

Table 5: Seed robustness across five complete walk-forward + backtest cycles. Sharpe CV = 0.025. IC (NO) = non-overlapping information coefficient.

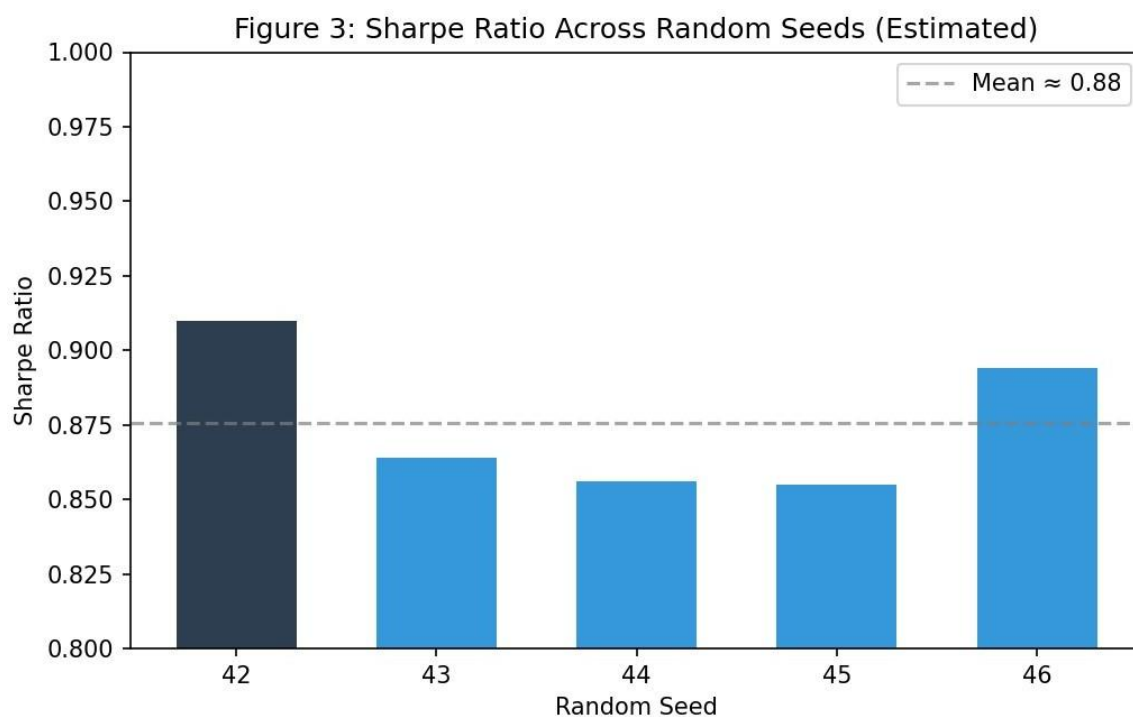


Figure 3: Sharpe ratio across five random seeds. The production seed (42) produces the lowest Sharpe. Cross-seed CV = 0.025.

The cross-seed Sharpe CV is 0.025, indicating that GMM and GBM randomness has minimal effect on the trading signal. The production seed (42) produces the lowest Sharpe in the set, meaning the reported results are conservative. Non-overlapping IC ranges from 0.065 to 0.114, consistently below significance thresholds regardless of seed.

6.2 Parameter Sensitivity

Vol Target	Sharpe	CAGR	Max DD	Calmar	Trades	Avg Alloc
10%	0.808	6.1%	−22.7%	0.268	24	70.1%
12%	0.825	7.0%	−25.1%	0.279	24	79.2%
15% (prod)	0.828	8.0%	−27.0%	0.295	24	89.4%
18%	0.797	7.9%	−28.4%	0.280	24	93.1%
20%	0.795	8.0%	−29.0%	0.276	24	94.0%
25%	0.798	8.1%	−30.2%	0.269	24	95.0%
Fixed 95%	0.794	8.1%	−30.9%	0.263	24	95.2%

Table 6: Volatility target sensitivity (golden cross + vol targeting only, without ML forecast exits; 24 trades). Sharpe CV = 0.017 across all targets. Cooldown sweep (0–15 days) equally flat: Sharpe 0.828–0.860. Note: the 24 trades and higher Sharpe (0.828) compared to Table 1’s golden cross + vol targeting row (0.699, 12 trades) reflect the vol-targeting sweep’s use of position re-entry after cooldown periods within the same golden cross regime, whereas Table 1’s configuration enters once per golden cross activation and holds until death cross.

The Sharpe is virtually invariant across volatility targets ($CV = 0.017$) and cooldown settings. The trade-off is transparent: lower vol targets compress drawdowns (10% target: −22.7% max DD) at the cost of CAGR (6.1% vs 8.1%). The system is not sensitive to either free parameter, consistent with the finding that its edge derives from the trend filter and exit asymmetry rather than from precise parameter calibration.

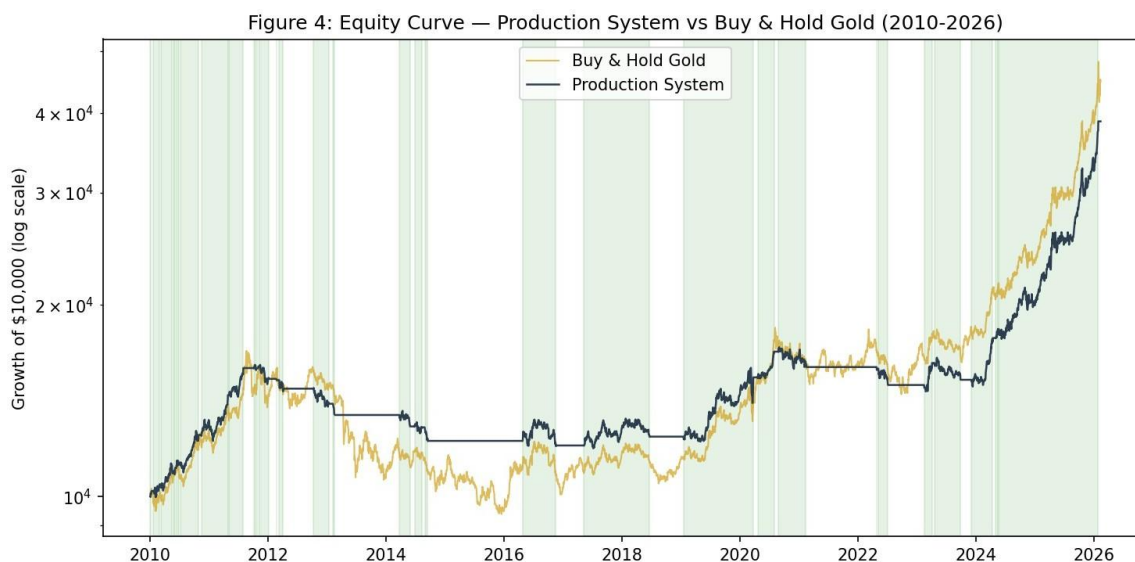


Figure 4: Equity curve of the corrected production system versus buy-and-hold gold, 2010–2026. Green shading indicates periods in position. Log scale.

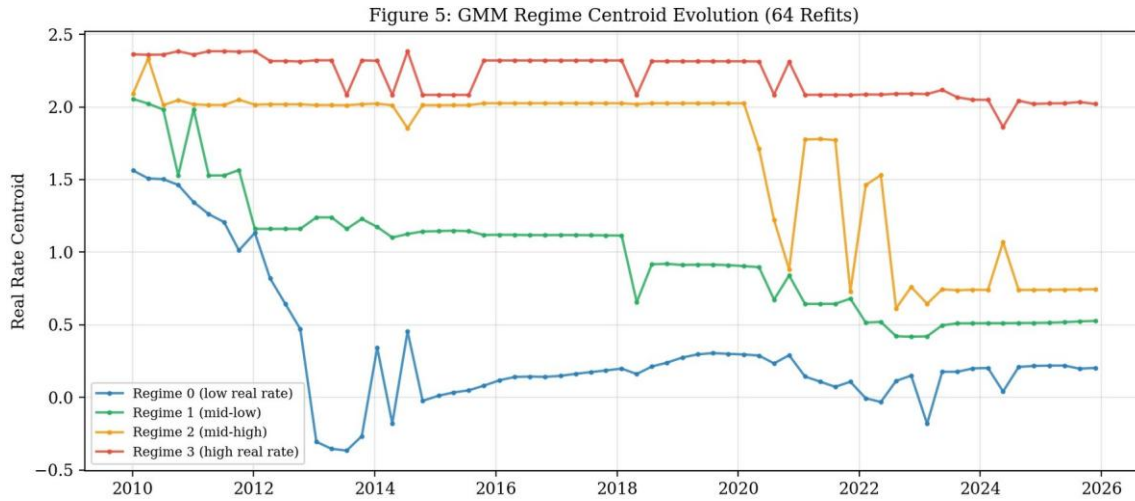


Figure 5: GMM regime centroid evolution (real rate dimension) across 64 quarterly refits. Regime 3 (high real rate, latest centroid 2.02) is the most stable; Regime 2 (centroid 0.75) shows the most drift.

The regime structure exhibits moderate stability over 64 refits (Figure 5), with three instances of adjacent centroid overlap. The OOS period shows 115 regime transitions with a median duration of 35 days. Given the insignificance of the forecast signal, this instability has negligible operational impact.

7. Discussion

7.1 The Complexity Question

The decomposition raises a direct question: is the ML stack justified? The golden cross alone achieves Sharpe 0.698 with zero parameters and no data dependencies beyond gold prices. The full system adds 0.21 Sharpe (to 0.910) at the cost of: a GMM with 4 regimes, four supervised models retrained quarterly, 15 features from 7 data sources, volatility-targeted sizing with two parameters, and a forecast-driven exit threshold. This complexity adds data pipeline risk, model monitoring requirements, and operational overhead.

The honest answer is nuanced. The CAGR modestly improves (8.4% to 8.8%), and drawdowns compress (−32.8% to −26.1%, a 6.7 percentage point reduction) through the exit asymmetry documented in Section 4.3. Whether this justifies the complexity depends on the investor's loss aversion and operational capability. For an institutional allocator, the answer may be yes. For an individual trader, the golden cross alone may be the better risk-adjusted choice when operational complexity is factored in.

7.2 What the ML Layer Actually Contributes

The evidence from the decomposition points to a specific and modest function: the ML layer provides noisy profit-taking that is not worse than random, combined with volatility-targeted position sizing. The forecast is insignificant (Section 4.2), the null model cannot distinguish the production system from random timing under uniform entry (Section 4.5), and a trailing stop captures 76% of the Sharpe with no ML infrastructure (Section 4.8).

What the ML layer does, concretely, is generate exit signals during extended positions. Because the −1.5% threshold fires when feature conditions deteriorate (rising real rates, falling momentum, weakening positioning), the exits tend to coincide with periods of extended strength, capturing profits before a trend reversal that the death cross would eventually catch anyway. The exit asymmetry (+10.4% for forecast exits vs −3.9% for death cross exits) is descriptively real but not statistically distinguishable from what random timing within the golden cross regime achieves.

This is a coherent but modest function. The ML layer is not predicting returns. It is not timing entries. It is generating noisy exit signals that happen to produce a positive asymmetry in combination with the golden cross's trend filter. Whether this function justifies four-model operational complexity is a practical question that depends on the implementation context.

7.3 The Look-Ahead Lesson

The training label leak illustrates a class of look-ahead bias that is distinct from the more commonly discussed feature-level look-ahead. In feature look-ahead, future data contaminates the model inputs. In label look-ahead, the inputs are clean but the training targets reference unobserved future prices. The distinction matters because researchers routinely audit features for look-ahead while assuming that the target variable, by definition forward-looking, cannot create leakage in a walk-forward framework. This assumption is wrong when the forecast horizon is shorter than the distance between the latest training row and the prediction date.

The practical recommendation is straightforward: for any forecast horizon H in trading days, embargo the last H rows of training data at each walk-forward refit. The cost is negligible (21 rows out of 1,500+), and the protection is substantial in this case, the difference between a statistically significant and insignificant forecast.

7.4 Limitations

Several limitations should be acknowledged. First, 30 trades over 16 years provide limited statistical power. The trade-level Sharpe CI spans $[0.15, 1.03]$, wide enough that neither the null model (Section 4.5) nor the Ledoit-Wolf test (Section 4.4) can confirm the ML layer adds value at conventional significance levels. The daily-level CIs $[0.42, 1.40]$ overlap with all benchmarks.

Second, the COT release lag correction was discovered during auditing, not pre-specified. Three of 15 forecast features derive from CFTC reports that cover Tuesday positions but release Friday. A 3-business-day lag was applied to all COT dates before merging. This correction improved performance (Sharpe from 0.83 to 0.91 in combination with the embargo), which constitutes another implicit comparison on the OOS period. The lag is the methodologically correct choice regardless of its performance impact, but corrections that happen to improve performance deserve extra skepticism, even when methodologically necessary. Had the correction hurt performance, it would still have been the right thing to do, but the fact that it went the favourable way should give pause.

Third, this study examines a single asset. The training label mechanism generalises theoretically to any walk-forward system with multi-period targets, but the alpha decomposition findings, specifically that the trend filter dominates and the ML forecast adds marginal value, may not hold in markets with different trend characteristics. In equity indices, where trends are weaker and mean-reversion is more common, the golden cross may contribute less. In other trending markets (crude oil, silver), the exit asymmetry mechanism might behave similarly, as any system that exits during extended trends and cuts losses on reversals will produce the same asymmetric profile. Testing across asset classes is the natural extension.

Fourth, FRED macroeconomic data were downloaded as latest-vintage rather than point-in-time, introducing a minor look-ahead through data revisions. This primarily affects CPI-derived features; real yield and breakeven inflation revisions are typically small (less than 5bp), but CPI seasonal adjustment revisions can be 20–30bp, and since CPI_YoY is a GMM regime feature, the propagation path is: CPI revision $\rightarrow$ regime misclassification $\rightarrow$ different forecast $\rightarrow$ different trades. The net impact is likely small given the GMM uses 7 features, but it is not negligible, but real-time replication would require a point-in-time data source such as ALFRED. Fifth, the decomposition treats layers as sequential, but interactions exist: the forecast-driven exit timing works differently without the golden cross providing the baseline trend filter. The layers are not fully independent.

Sixth, and most importantly, the system was developed iteratively on the same OOS period. Earlier architectures were tested and discarded; feature sets were pruned; thresholds were adjusted. Each of these decisions used information from the 2010–2026 period, creating an implicit multiple comparisons problem that the training embargo cannot fix. The embargo corrects a specific, identifiable bias in the walk-forward mechanics. It does not address the broader selection bias from iterative development. The trade concentration analysis (Section 4.9) and the first-half Sharpe of 0.44 provide a partial check: the system's relative advantage over

buy-and-hold persists in both sub-periods, suggesting the edge is not entirely a product of one favorable realization. But the absolute performance figures should be interpreted with this development history in mind.

8. Conclusion

This paper decomposes the alpha of a four-layer gold timing system and finds that the trend filter provides the structural foundation, while the ML forecasting layer does not reliably add value beyond random timing within the trend regime. The forecast is insignificant, the null model results are ambiguous ($p = 0.01$ – 0.26 depending on entry assumptions), and a simple trailing stop captures 76% of the full system's Sharpe with no ML infrastructure. The evidence is genuinely ambiguous: the ML layer's contribution is real but imprecisely estimated, and its magnitude depends on the baseline assumption.

The ML layer's residual contribution is noisy profit-taking that happens to produce a positive exit asymmetry, but one that is not statistically distinguishable from what random timing achieves. This is a coherent but modest function that may not justify the operational complexity of a four-model ensemble with quarterly retraining. Performance is heavily concentrated: the three largest trades account for 82% of total returns, and the first-half Sharpe (2010–2017) is 0.44, suggesting that the full-period figure of 0.91 is substantially dependent on the 2024–2026 gold rally.

This honest attribution required discovering that training label leakage had inflated the forecast's apparent quality, transforming an insignificant signal into an apparently significant one (IC 0.08 $\rightarrow$ 0.23, $p = 0.001$). Correcting it with a 21-day embargo and 3-day COT release lag revealed the system's true alpha sources.

For practitioners building multi-layer trading systems, three lessons emerge. First, decompose alpha across layers and test each against simple alternatives before trusting headline performance. Second, apply an H-day embargo to walk-forward training when using multi-period forward targets, and lag any features with known publication delays. Third, report forecast quality metrics (IC, directional accuracy) alongside backtest metrics (Sharpe, CAGR). A system with high Sharpe and low IC has structural alpha, not predictive alpha—a distinction that matters for monitoring, maintenance, and expectations.

These findings are likely most relevant to trending, macro-sensitive assets like gold, where moving-average filters capture persistent regime shifts driven by real rate and inflation dynamics. In markets with weaker trend structure, the golden cross may contribute less, and the relative importance of the ML layer could differ. Testing across asset classes is the natural extension.

As of the final data point in the sample (February 10, 2026), the system signals EXIT with a REGIME ENS forecast of -4.75% in Regime 3 (high real rates), with gold at \$5,023 and 30% above its 200-day moving average. This is reported for replicability, not as a market call.

Data Availability

All data used in this study are freely available from public sources. Gold (XAU/USD) spot prices are sourced from Investing.com. Macroeconomic variables (DFII10, T10YIE, VIXCLS, DFF,

BAMLH0A0HYM2, DTWEXBGS) are obtained from the Federal Reserve Economic Data (FRED) database via their public API using the exact series IDs listed above. Commitments of Traders data are from the CFTC weekly Futures-Only reports for gold (commodity code 088691).

Appendix A: Feature Definitions

Table A1 lists all features used in the regime detection and forecasting stages. Regime features (R) are inputs to the GMM. Forecast features (F) are inputs to the supervised models. Some features appear in both stages.

Feature	Stage	Source	Construction
Real_Rate_10Y	R, F	FRED (DFII10)	10-year TIPS real yield, daily
VIX	R, F	FRED (VIXCLS)	CBOE Volatility Index, daily
Yield_Curve	R, F	FRED (10Y-2Y)	10-year minus 2-year Treasury yield
HY_Spread	R	FRED	BofA high-yield option-adjusted spread (BAMLH0A0HYM2)
CPI_YoY	R	FRED	CPI year-over-year change, monthly (forward-filled)
Dollar_Ret_3M	R	FRED	63-day return of trade-weighted dollar index (DTWEXBGS)
Gold_Vol_60	R	Derived	60-day rolling annualised volatility of gold returns
Gold_Ret_1M	F	Derived	21-day gold return
Gold_Ret_3M	F	Derived	63-day gold return
Gold_Vol_Ratio	F	Derived	Ratio of 20-day to 60-day gold volatility
Gold_MA_Ratio	F	Derived	Ratio of 50-day to 200-day gold moving average
Breakeven_10Y	F	FRED (T10YIE)	10-year breakeven inflation rate, daily
Fed_Rate	F	FRED (DFF)	Effective federal funds rate, daily
ATR_Ratio	F	Derived	Ratio of 20-day to 60-day Average True Range
Avg_Range_20	F	Derived	20-day mean of (High-Low)/Close
COT_MM_Net_Z52	F	CFTC	Money manager net positioning, z-scored over 52 weeks
COT_GoldSilver_Spread	F	CFTC	Gold minus silver net positioning spread
COT_Conc4_Net	F	CFTC	Net concentration ratio of top 4 traders
Dollar_Ret_1M	F	FRED	21-day return of trade-weighted dollar index

Table A1: Feature definitions. R = regime detection (GMM input), F = forecasting (supervised model input). All COT features are lagged 3 business days to account for the Tuesday-to-Friday publication delay. Target variable: 21-day forward gold return (Fwd_1M).

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